

Database System Engineering and Distributed Backend Development

WEEK-1 : BOOK_FLOW

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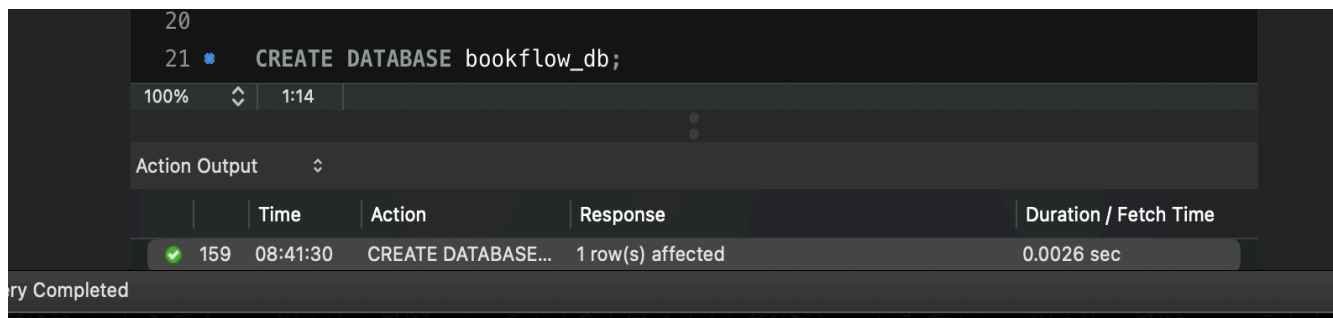
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Section: 7

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Step 1 — Create Database

```
CREATE DATABASE IF NOT EXISTS bookflow_db;
```



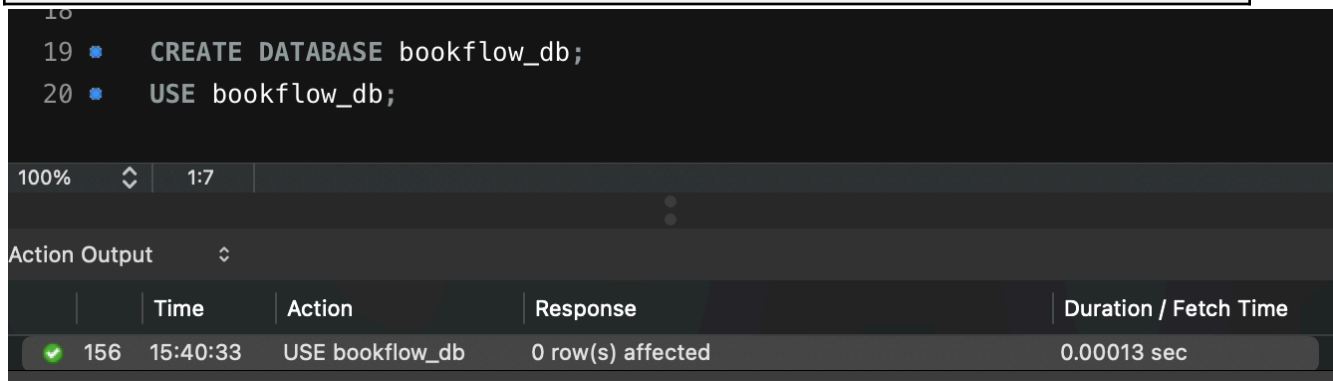
The screenshot shows a database client interface with a dark theme. The command 'CREATE DATABASE bookflow_db;' is entered in the top text area. Below it, a progress bar indicates 100% completion. The 'Action Output' section is expanded, showing a table with columns: Time, Action, Response, and Duration / Fetch Time. A single row is displayed with a green checkmark icon, indicating successful execution.

	Time	Action	Response	Duration / Fetch Time
✓ 159	08:41:30	CREATE DATABASE...	1 row(s) affected	0.0026 sec

Query Completed

Step 2 — Select the Database

```
USE bookflow_db;
```

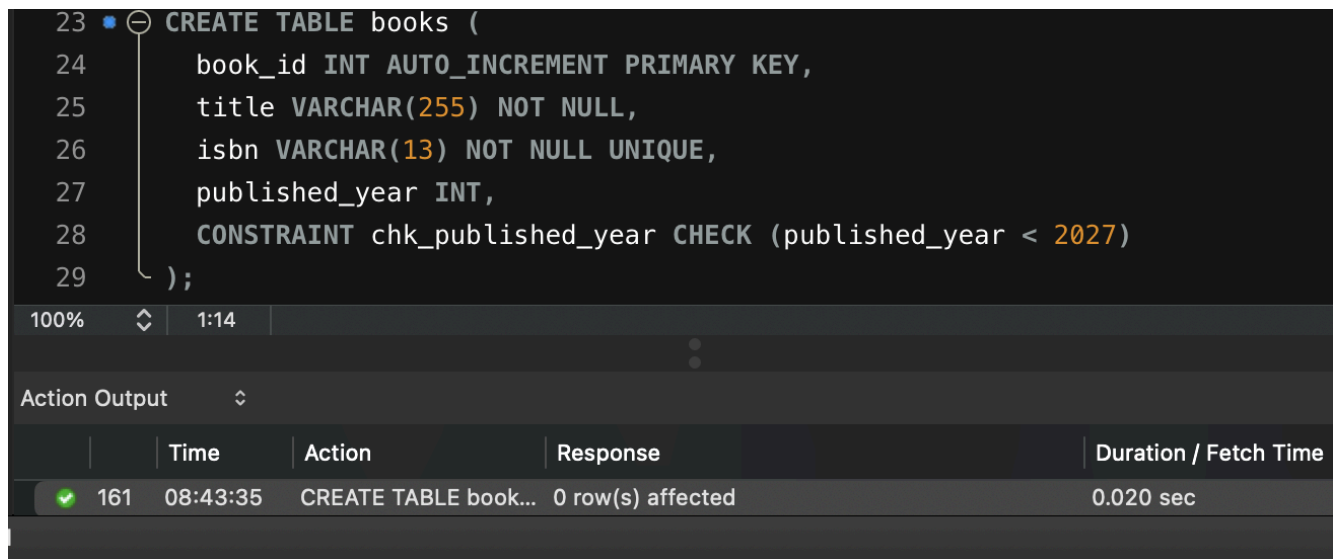


The screenshot shows the same database client interface. The command 'USE bookflow_db;' is entered in the top text area. The 'Action Output' section is expanded, showing a table with columns: Time, Action, Response, and Duration / Fetch Time. A single row is displayed with a green checkmark icon, indicating successful execution.

	Time	Action	Response	Duration / Fetch Time
✓ 156	15:40:33	USE bookflow_db	0 row(s) affected	0.00013 sec

Step 3 — Create Books Table (DDL + Constraints)

```
CREATE TABLE books (  
  book_id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(255) NOT NULL,  
  isbn VARCHAR(13) NOT NULL UNIQUE,  
  published_year INT,  
  CONSTRAINT chk_published_year CHECK (published_year < 2027)  
);
```



The screenshot shows a SQL IDE with a dark theme. The SQL editor contains the following code:

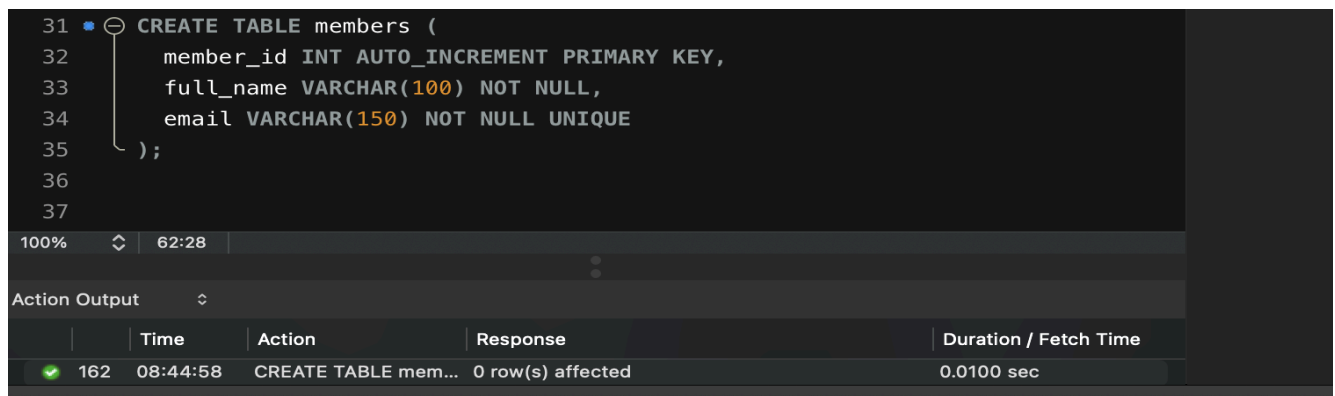
```
23 CREATE TABLE books (  
24     book_id INT AUTO_INCREMENT PRIMARY KEY,  
25     title VARCHAR(255) NOT NULL,  
26     isbn VARCHAR(13) NOT NULL UNIQUE,  
27     published_year INT,  
28     CONSTRAINT chk_published_year CHECK (published_year < 2027)  
29 );
```

Below the editor, the 'Action Output' tab is active, displaying a table with the following data:

	Time	Action	Response	Duration / Fetch Time
✓ 161	08:43:35	CREATE TABLE book...	0 row(s) affected	0.020 sec

Step 4 — Create Members Table

```
CREATE TABLE members (  
  member_id INT AUTO_INCREMENT PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(150) NOT NULL UNIQUE  
);
```



The screenshot shows a SQL IDE with a dark theme. The SQL editor contains the following code:

```
31 CREATE TABLE members (  
32     member_id INT AUTO_INCREMENT PRIMARY KEY,  
33     full_name VARCHAR(100) NOT NULL,  
34     email VARCHAR(150) NOT NULL UNIQUE  
35 );  
36  
37
```

Below the editor, the 'Action Output' tab is active, displaying a table with the following data:

	Time	Action	Response	Duration / Fetch Time
✓ 162	08:44:58	CREATE TABLE mem...	0 row(s) affected	0.0100 sec

Step 5 — Verify Table Structures

```
DESCRIBE books;
```

36
37 DESCRIBE books;
38

100% 1:30

Result Grid Filter Rows: Search Export:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(255)	NO		NULL	
isbn	varchar(13)	NO	UNI	NULL	
published_year	int	YES		NULL	

Result 1 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 164	08:46:29	DESCRIBE books	4 row(s) returned	0.016 sec / 0.000015...

```
DESCRIBE members;
```

38
39 DESCRIBE members;
40

100% 25:33

Result Grid Filter Rows: Search Export:

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Form Editor

Step 6 — Insert 3 Books and Display

```
INSERT INTO books (title, isbn, published_year) VALUES
('Ponniyin Selvan', '9788172234809', 1955),
('The White Tiger', '9781416562603', 2008),
('Malgudi Days', '9780143039655', 1943);
```

```
40
41 INSERT INTO books (title, isbn, published_year) VALUES
42 ('Ponniyin Selvan', '9788172234809', 1955),
43 ('The White Tiger', '9781416562603', 2008),
44 ('Malgudi Days', '9780143039655', 1943);
45
```

100% 3:35

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 166	08:47:52	INSERT INTO books...	3 row(s) affected Records: 3 Duplicates: 0 Wa...	0.0055 sec

```
SELECT * FROM books;
```

```
47 SELECT * FROM books;
48
```

100% 1:40

Result Grid Filter Rows: Search Edit:

	book_id	title	isbn	published_ye...	
	1	Ponniyin Selvan	9788172234809	1955	
	2	The White Tiger	9781416562603	2008	
	3	Malgudi Days	9780143039655	1943	
	NULL	NULL	NULL	NULL	

Step 7 — Insert 3 Members and Display

```
INSERT INTO members (full_name, email) VALUES
('Hansika Reddy', 'gunapatihansikareddy@klh.edu.in'),
('Arjun Rao', 'arjun.rao@example.com'),
('Sneha Iyer', 'sneha.iyer@example.com');
```

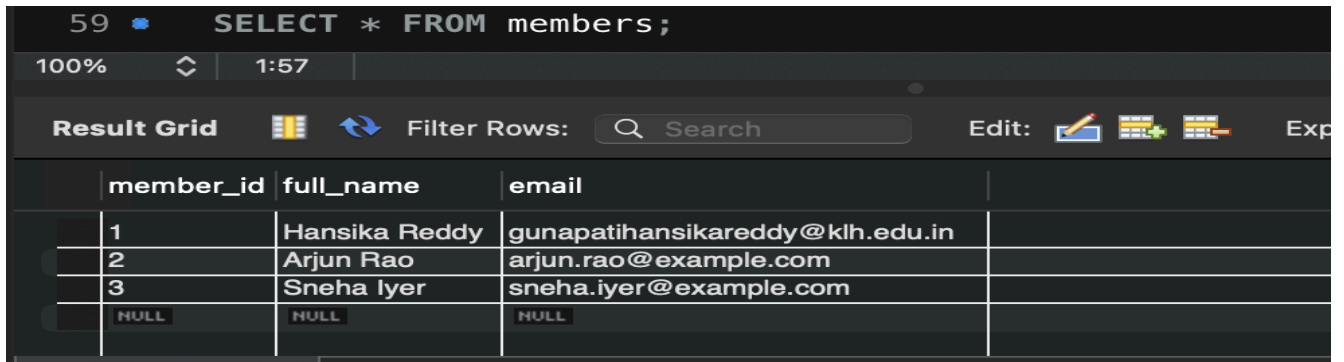
```
50
51 INSERT INTO members (full_name, email) VALUES
52 ('Hansika Reddy', 'gunapatihansikareddy@klh.edu.in'),
53 ('Arjun Rao', 'arjun.rao@example.com'),
54 ('Sneha Iyer', 'sneha.iyer@example.com');
55
```

100% 21:47

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 168	08:49:23	INSERT INTO memb...	3 row(s) affected Records: 3 Duplicates: 0 Wa...	0.0021 sec

```
SELECT * FROM members;
```



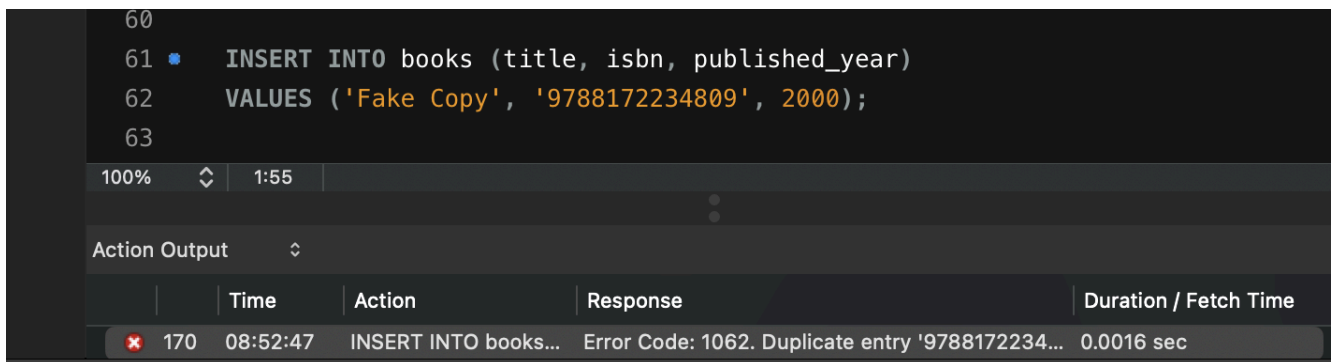
The screenshot shows a database client interface with a query editor at the top containing the SQL statement `SELECT * FROM members;`. Below the editor, a toolbar includes options for 'Result Grid', 'Filter Rows', and a search bar. The main area displays a table with the following data:

	member_id	full_name	email
	1	Hansika Reddy	gunapatihansikareddy@klh.edu.in
	2	Arjun Rao	arjun.rao@example.com
	3	Sneha Iyer	sneha.iyer@example.com
	NULL	NULL	NULL

Step 8 — Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9788172234809', 2000);
```



The screenshot shows a database client interface with a query editor containing the SQL statement:

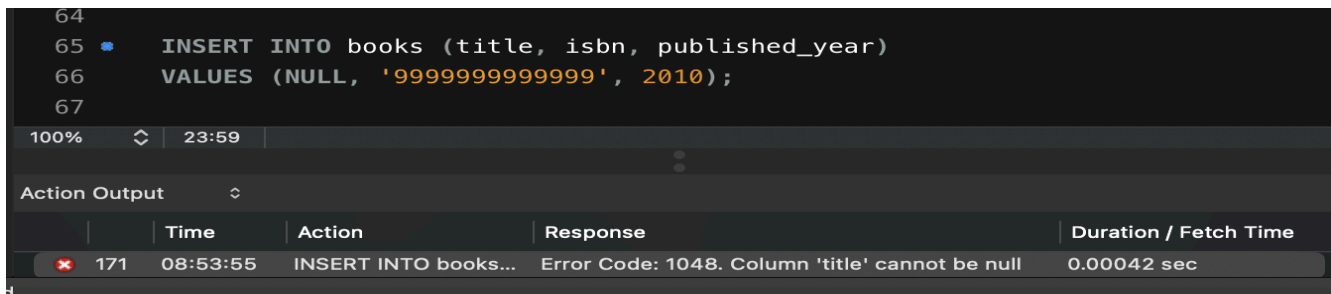
```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9788172234809', 2000);
```

Below the editor, the 'Action Output' section shows a failed execution:

	Time	Action	Response	Duration / Fetch Time
✖ 170	08:52:47	INSERT INTO books...	Error Code: 1062. Duplicate entry '9788172234...	0.0016 sec

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```



The screenshot shows a database client interface with a query editor containing the SQL statement:

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Below the editor, the 'Action Output' section shows a failed execution:

	Time	Action	Response	Duration / Fetch Time
✖ 171	08:53:55	INSERT INTO books...	Error Code: 1048. Column 'title' cannot be null	0.00042 sec

Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '88888888888888', 2030);
```

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70
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INSERT INTO books (title, isbn, published_year)

VALUES ('Time Traveler', '888888888888', 2030);

100%

43:61

Action Output

	Time	Action	Response	Duration / Fetch Time
✖ 172	08:54:57	INSERT INTO books...	Error Code: 3819. Check constraint 'chk_publis...	0.0012 sec

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Duplicate Person', 'gunapatihansikareddy@klh.edu.in');
```

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74
75
76

INSERT INTO members (full_name, email)

VALUES ('Duplicate Person', 'gunapatihansikareddy@klh.edu.in');

100%

43:61

Action Output

	Time	Action	Response	Duration / Fetch Time
✖ 173	08:56:46	INSERT INTO memb...	Error Code: 1062. Duplicate entry 'gunapatihan...	0.0011 sec